

CSE 4631

# Digital Signal Processing

Week 11

# Applications of the DFT

- Calculation of a signal's *frequency spectrum* – direct examination of information encoded in the frequency, phase, and amplitude of the component sinusoids.
- Finding a system's frequency response from the system's impulse response, and vice versa, allowing the system to be analyzed in the frequency domain. (just as convolution does in the time domain)
- Acting as an intermediate step in more elaborate signal processing techniques, such as FFT convolution.

# Spectral Analysis of Signals

- It is very common for information to be encoded in the sinusoids that form a signal
- Many things oscillate in our universe:
  - Speech – result of vibration of the human vocal cords
  - Stars and planets change their brightness as they rotate on their axes and revolve around each other
  - Ship's propellers generate periodic displacement of the water, etc.
- The *frequency, phase, and amplitude* of the component sinusoids of these signals contain the key information. (The *shape* of the time domain waveform is not important)
- DFT is used to extract these information

# Spectral Analysis of Signals - Example: Underwater Sound

Microphone is placed in the water

Resulting electronic signal is amplified

Analog low-pass filter is used to remove all frequencies above 80 hertz

Resulting signal is sampled 160 samples per second

Signal is multiplied by a smooth curve called a Hamming Window (b,c)

**Observation 1:** Noisy waveform conveying little information (a)

Let's consider 256 samples for this example's sake

Several thousands samples are acquired and stored

Samples near the ends have been reduced in amplitude

DFT

Conversion to polar notation

**Observation 2:** still looking messy. (d)  
And a longer DFT (e.g., 2048 point) doesn't help either

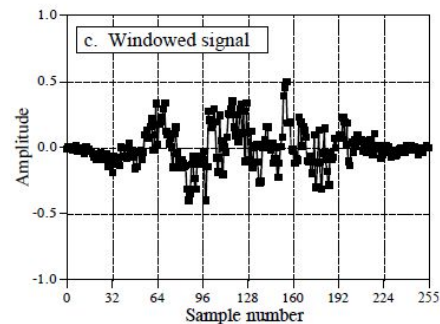
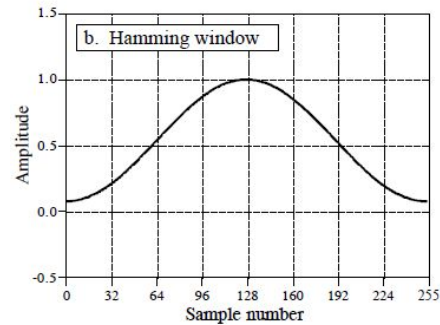
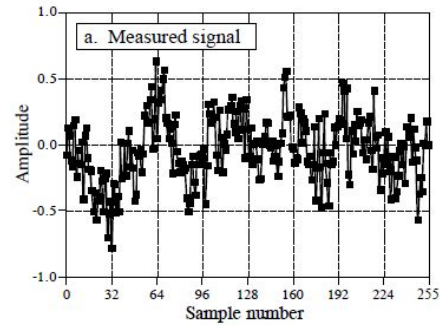
We need to use more of the original signal in a way that doesn't increase the number of points in the frequency spectrum

Even though the original 2048 points contain more information, the greater number of samples also dilutes them by the same factor

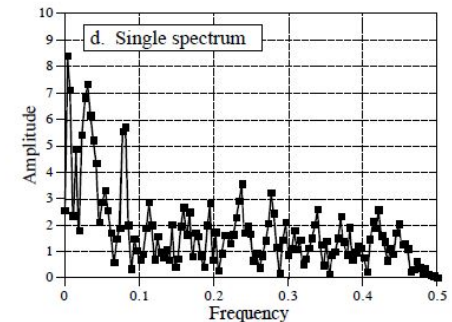
## Time Domain

## Frequency Domain

An example of spectral analysis. Figure (a) shows 256 samples taken from a (simulated) undersea microphone at a rate of 160 samples per second. This signal is multiplied by the Hamming window shown in (b), resulting in the windowed signal in (c). The frequency spectrum of the windowed signal is found using the DFT, and is displayed in (d) (magnitude only).



DFT



# Spectral Analysis of Signals - Example: Underwater Sound (contd.)

Break the input signal into 100 256 point segments

Follow the highlighted steps for these segments

Resulting frequency spectra are averaged (only the magnitude) to form a single 129 point frequency spectrum

**Observation 3:**  
Interesting features can be observed due to reduced noise level (e)

## Method 1

Some applications might average millions of segments to bring out weak features

Random noise reduces in proportion to the square-root of the number of segments

## Method 2

Take a very long DFT, e.g., 16,384 points

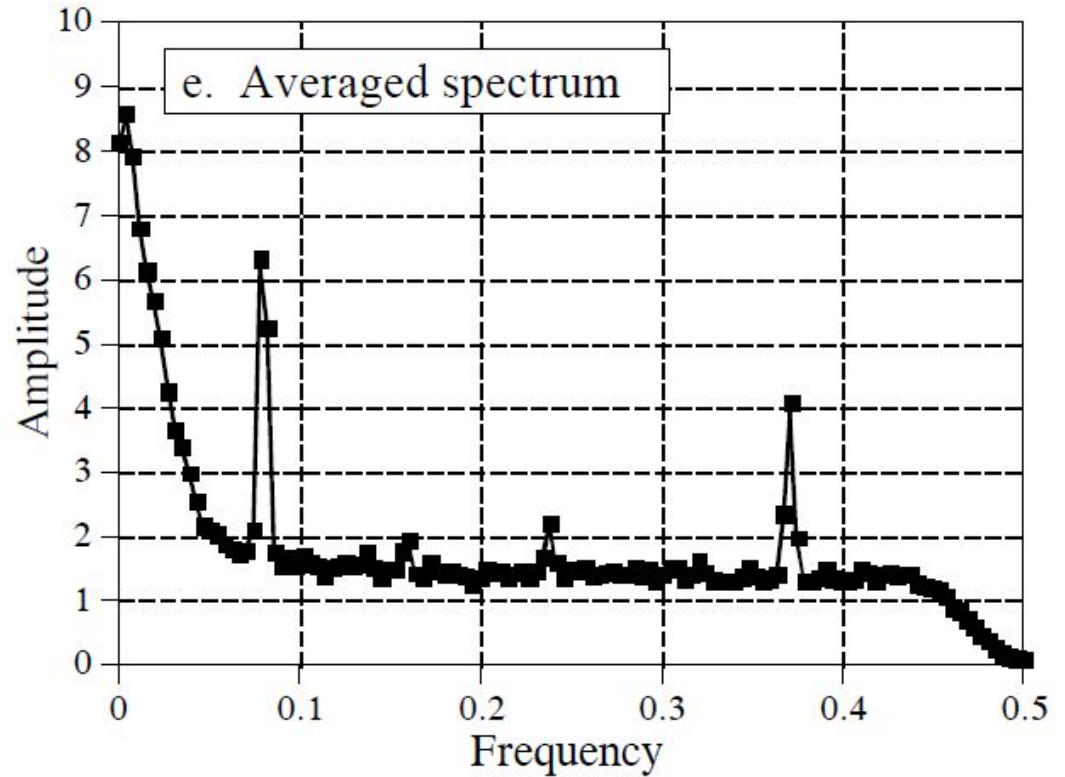
Resulting frequency is high resolution (8193 samples), but very noisy

A low-pass digital filter is then used to smooth the spectrum

Reduces the noise at the expense of the resolution

# Spectral Analysis of Signals - Example: Underwater Sound (contd.)

Averaging 100 of those (d) frequency spectra reduces the random noise, resulting in the averaged frequency spectrum shown in (e).



# Spectral Analysis of Signals - Example: Underwater Sound (contd.)

## Method 1

- Easier to use, because the digital filter isn't needed

## Method 2

- Has potential to perform better, because the digital filter can be tailored to optimize the trade-off between noise and resolution
- However, this improved performance is seldom worth the trouble. Both noise and resolution can be improved by using more data from the input signal.

Think of this:

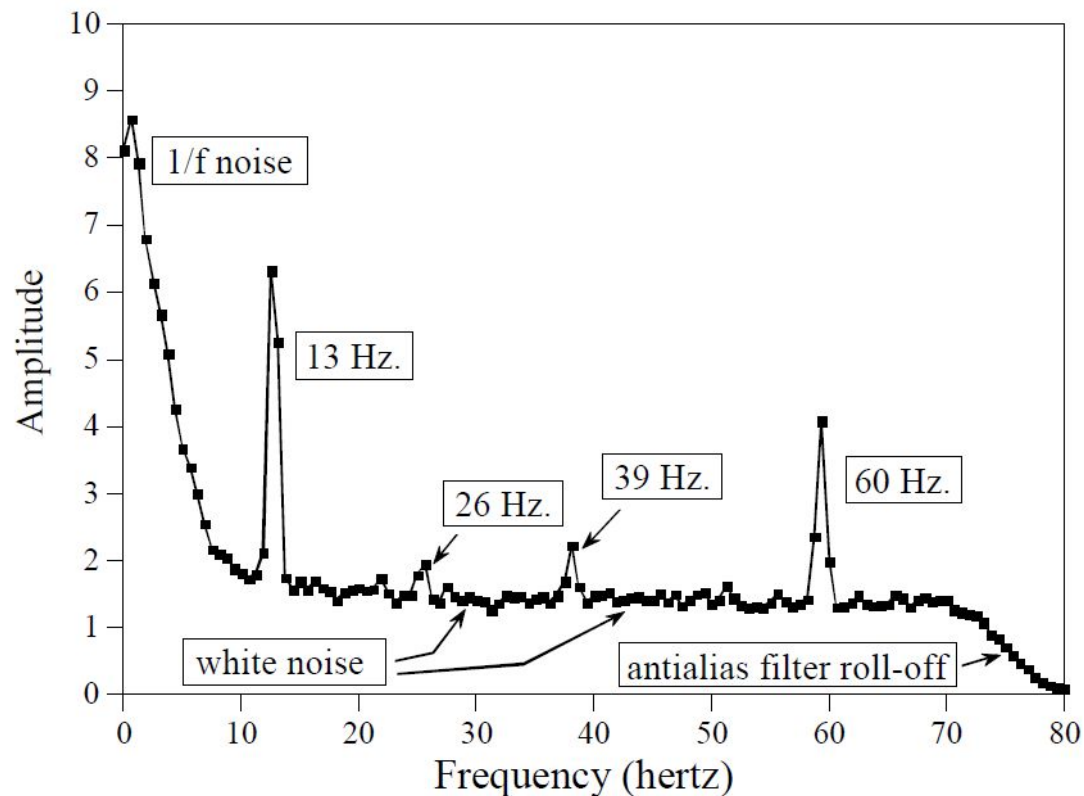
1,63,840,000 samples  $\longrightarrow$  10,000 segments of 16,384 samples each

Resulting frequency spectrum is high resolution (8193 points) and low noise (10,000 averages)



# Spectral Analysis of Signals - Example: Underwater Sound (contd.)

Example frequency spectrum. Three types of features appear in the spectra of acquired signals: (1) random noise, such as white noise and 1/f noise, (2) interfering signals from power lines, switching power supplies, radio and TV stations, microphonics, etc., and (3) real signals, usually appearing as a fundamental plus harmonics. This example spectrum (magnitude only) shows several of these features.



# Spectral Analysis of Signals (contd.)

- Suppose there are peaks very close together
- There are **two factors** that limit the frequency resolution that can be obtained, that is, how close the peaks can be without merging into a single entity

## Factor 1: The length of the DFT

To separate two closely spaced frequencies, the sample spacing must be smaller than the distance between the two peaks

## Factor 2: The length of the Signal

Imagine a signal is created by adding two sine waves with only a slight difference in their frequencies. Over a short segment of this signal (a few periods), the waveform will look like a *single* sine wave.

The closer the frequencies, the longer the segment must be to conclude that more than one frequency is present.

## Spectral Analysis of Signals (contd.)

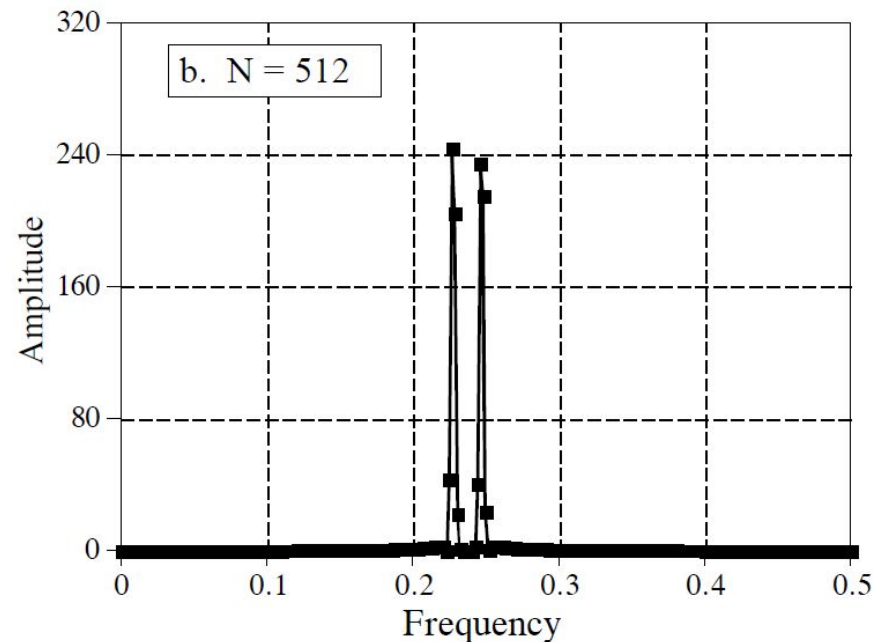
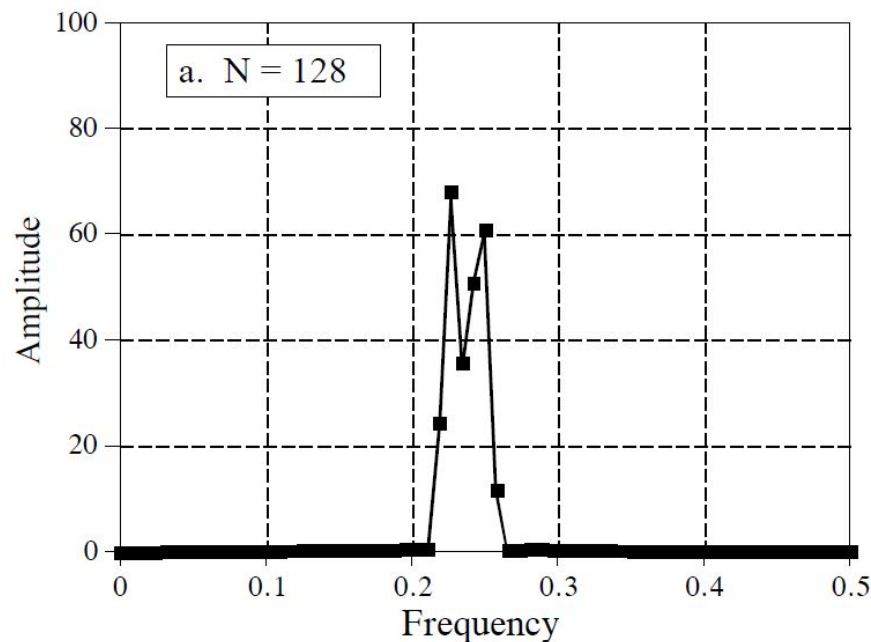


FIGURE 9-3

Frequency spectrum resolution. The longer the DFT, the better the ability to separate closely spaced features. In these example magnitudes, a 128 point DFT cannot resolve the two peaks, while a 512 point DFT can.

# Spectral Analysis of Signals (contd.)

**Question: How are the factors different?**

The length of the input signal does not have to be the same as the length of the DFT. You can add as many zeros as you like (padding) resulting in a longer DFT.

However, the added zeros won't change the shape of the spectrum, they only provide more samples in the frequency domain.

**Question: What happens if the input signal contains a sinusoid with a frequency between two of the basis functions?**

## Spectral Analysis of Signals (contd.)

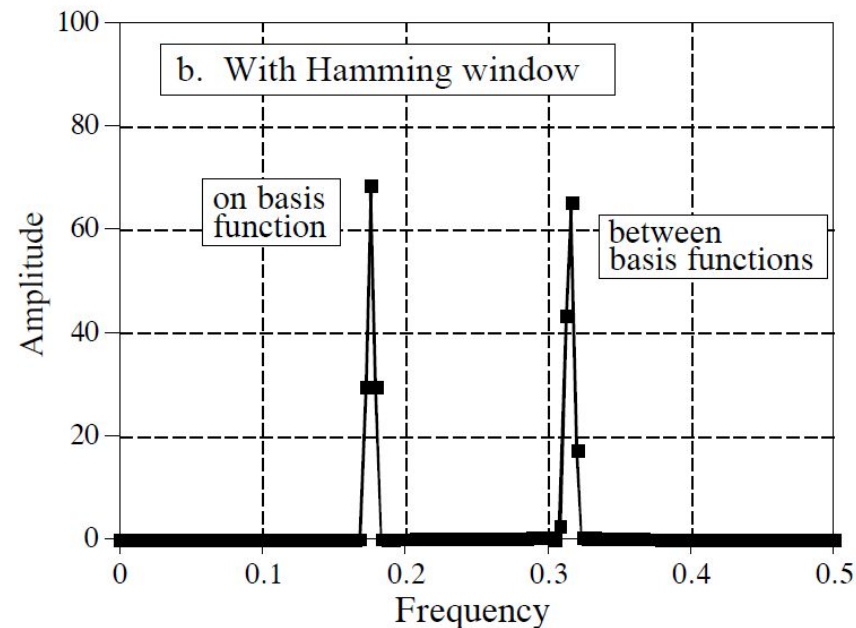
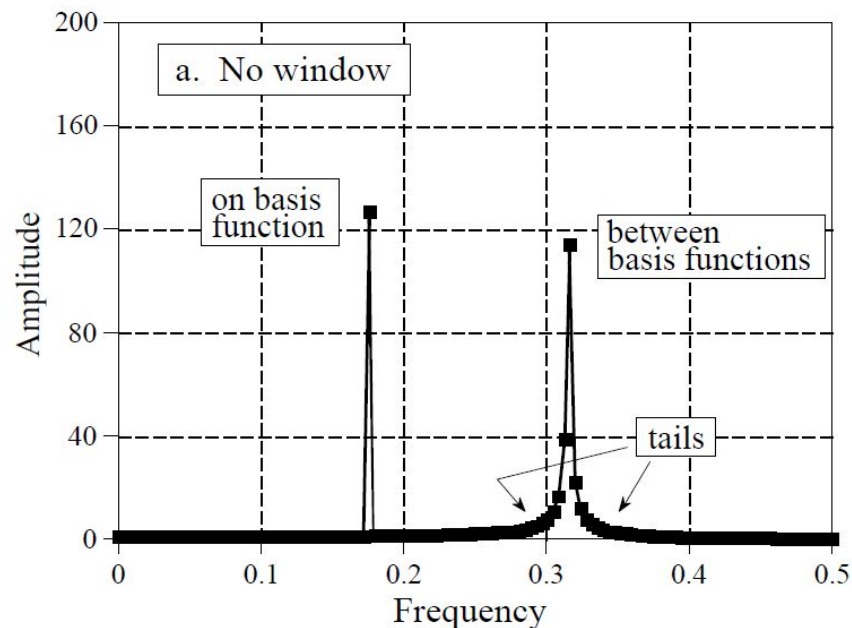


FIGURE 9-4

Example of using a window in spectral analysis. Figure (a) shows the frequency spectrum (magnitude only) of a signal consisting of two sine waves. One sine wave has a frequency exactly equal to a basis function, allowing it to be represented by a single sample. The other sine wave has a frequency *between* two of the basis functions, resulting in *tails* on the peak. Figure (b) shows the frequency spectrum of the same signal, but with a Hamming window applied before taking the DFT. The window makes the peaks look the same and reduces the tails, but broadens the peaks.

# Spectral Analysis of Signals (contd.)

Changes in the spectrum after using the window

- Makes the peaks look more alike – Good
- Reduces the amplitude of the tails – Good
- Reduces the resolution by broadening the peaks – Bad

In DSP jargon, windows provide a trade-off between *resolution* (the width of the peak) and *spectral leakage* (the amplitude of the tails)

# Frequency response of a system

- Systems are analyzed in time domain by convolution and same can be done in the frequency domain
- Any linear system can be completely described by how it changes the amplitude and phase of cosine waves passing through it – **Frequency response of the system**
- Since both the impulse response and the frequency response contain complete information about the system, there is a one-to-one correspondence between the two: Given one, you can calculate the other.
- **A system's frequency response is the Fourier Transform of its impulse response.**
- **Convolution in the time domain corresponds to multiplication in the frequency domain.**

$$x[n] * h[n] = y[n]$$

$$X[f] \times H[f] = Y[f]$$

## Frequency response of a system (contd.)

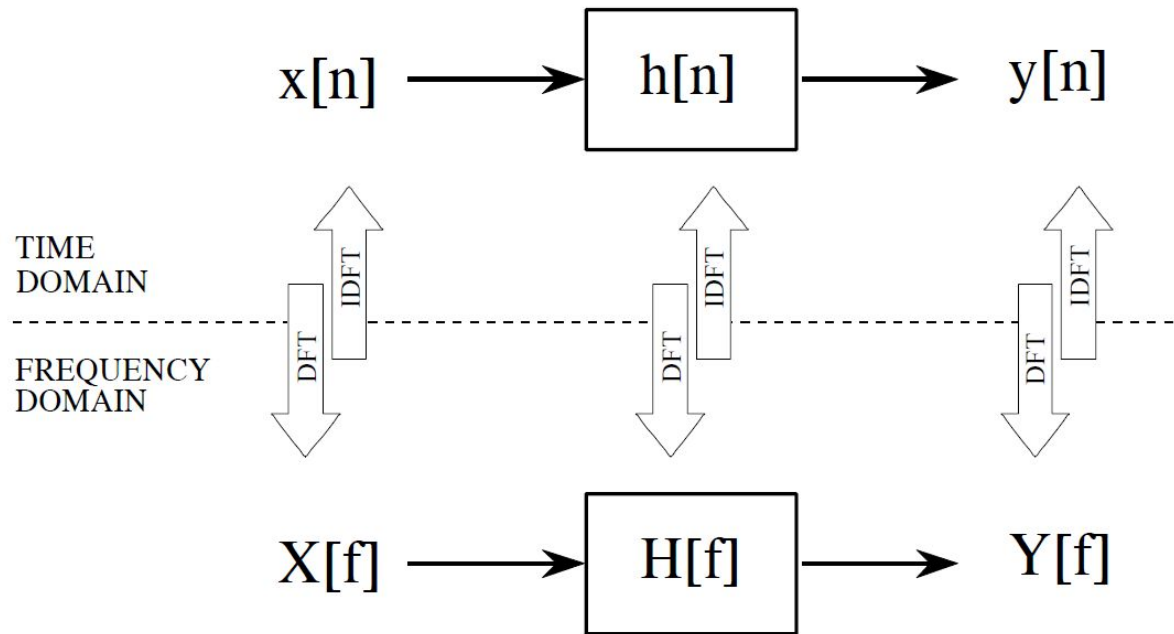


FIGURE 9-6

Comparing system operation in the time and frequency domains. In the time domain, an input signal is *convolved* with an impulse response, resulting in the output signal, that is,  $x[n] * h[n] = y[n]$ . In the frequency domain, an input spectrum is *multiplied* by a frequency response, resulting in the output spectrum, that is,  $X[f] \times H[f] = Y[f]$ . The DFT and the Inverse DFT relate the signals in the two domain.



## Frequency response of a system (contd.)

**Question: How much resolution can you obtain in the frequency response?**

Infinitely high, if you are willing to pad the impulse response with an infinite number of zeros. In other words, there is nothing limiting the frequency resolution except the length of the DFT.

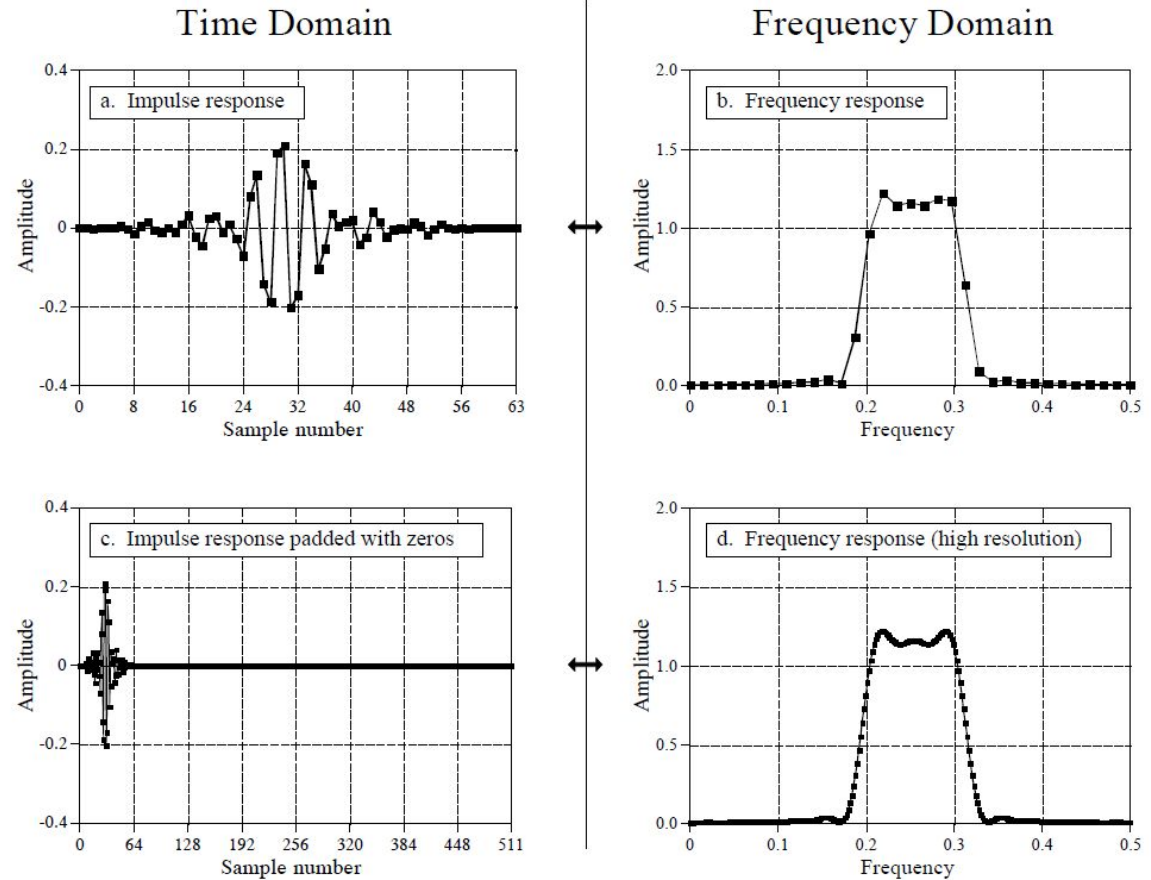


FIGURE 9-7

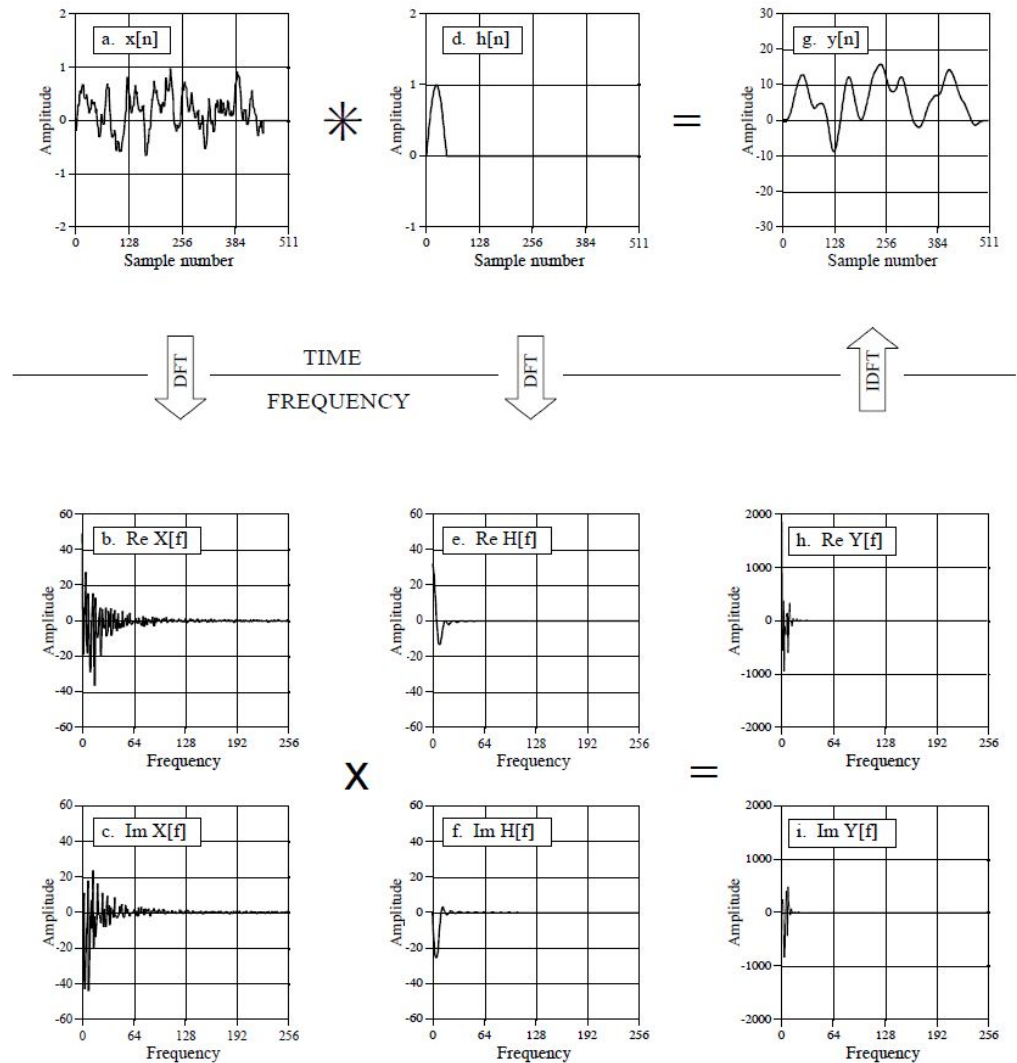
Finding the frequency response from the impulse response. By using the DFT, a system's impulse response, (a), can be transformed into the system's frequency response, (b). By padding the impulse response with zeros (c), higher resolution can be obtained in the frequency response, (d). Only the magnitude of the frequency response is shown in this example; discussion of the phase is postponed until the next chapter.

# Frequency response of a system (contd.)

## DFT vs DTFT for calculating the frequency response

- DFT considers the time domain signal to be *infinitely long* and *periodic*. That is, the  $N$  points are repeated over and over from negative to positive infinity.
- But if we pad the time domain signal with an ever increasing number of zeros, we can obtain a finer and finer sampling in the frequency domain.
- If we add an infinite number of zeros to the time domain signal, we have an *aperiodic* signal which has a frequency domain with infinitesimally small spacing between samples – as if it has become a *continuous signal*.
- Discrete Aperiodic signal Continuous curve DTFT!
- Therefore, in mathematical terms, a system's frequency response is found by taking the DTFT of its impulse response.
- Since this cannot be done in a computer, the DFT is used to calculate a sampling of the true frequency response.

# Convolution via the Frequency Domain



**Note:** (a) contains 453 nonzero samples, and (b) contains 60 nonzero samples. That's why the convolution of the two shown in (c) can fit comfortably in the 512 points provided

# Convolution via the Frequency Domain (contd.)

Convolution is avoided for **two reasons**:

- Mathematically difficult to deal with
  - Deconvolution is virtually impossible to understand in the time domain
  - Deconvolution can be carried out in the frequency domain as a simple division
  - The frequency domain becomes attractive whenever the complexity of the Fourier Transform is less than the complexity of the convolution
- Computation Speed
  - The standard convolution algorithm is slow since it requires a large number of multiplications and additions.
  - Does bringing the problem into the frequency domain via the DFT help? No. Just as many calculations will be required to calculate the DFTs, as are required to directly calculate the convolution. Solution? FFT! (we'll get back to it later)

# Convolution via the Frequency Domain (contd.)

**What does it mean to write:  $X[f] \times H[f] = Y[f]$**

In polar notation:

- Magnitudes are multiplied:  $MagY[f] = MagX[f] \times MagH[f]$
- Phases are added:  $PhaseY[f] = PhaseX[f] + PhaseH[f]$

} The polar form of the frequency response directly describes how the amplitudes and phases of the two signals (input and output) are related

In rectangular notation:

- $ReY[f] = ReX[f] ReH[f] - ImX[f] ImH[f]$
- $ImY[f] = ImX[f] ReH[f] + ReX[f] ImH[f]$

} These equations result when the operations above are carried out in the rectangular form.

# Convolution via the Frequency Domain (contd.)

**What does it mean to write:  $H[f] = Y[f] / X[f]$**

In polar notation:

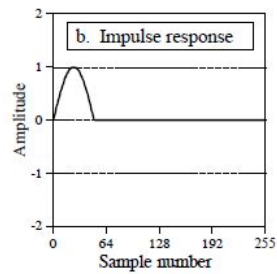
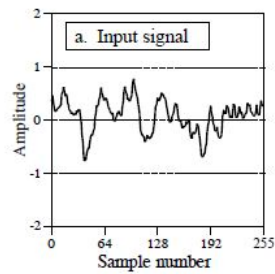
- Magnitudes are divided:  $MagH[f] = MagY[f] / MagX[f]$
- Phases are subtracted:  $PhaseH[f] = PhaseY[f] - PhaseX[f]$

In rectangular notation:

$$\left. \begin{aligned} ReH[f] &= \frac{ReY[f] ReX[f] + ImY[f] ImX[f]}{ReX[f]^2 + ImX[f]^2} \end{aligned} \right\} \begin{array}{l} \text{Multiply first eqn by } ReX \text{ and second eqn by } ImX \\ \text{and add them} \end{array}$$
$$\left. \begin{aligned} ImH[f] &= \frac{ImY[f] ReX[f] - ReY[f] ImX[f]}{ReX[f]^2 + ImX[f]^2} \end{aligned} \right\} \begin{array}{l} \text{Multiply first eqn by } ImX \text{ and second eqn by } ReX \\ \text{and add them} \end{array}$$

# Convolution via the Frequency Domain (contd.)

- The input signal is 256 points long.
- The impulse response 51 points long.
- Hence, the output is 306 points long.
- If we use 256 point DFTs to move the input and impulse into the frequency domain, after the multiplication, a 256 point inverse DFT will be used to find the output signal.
- How do you squeeze 306 values of the correct output signal into the 256 points? You can't.
- $N$  point DFT views the time domain as being an indefinitely long periodic signal, with  $N$  samples per period.
- Therefore, frequency domain tries to place the 306 point correct output signal into each of these 256 point periods by **pushing 49 of the samples into the neighboring period** to the right where they overlap with the samples that are legitimately there – **Circular Convolution!**
- **How do you overcome this?** Just pad each of the signals being convolved with enough zeros to allow the output signal room to handle the  $N+M-1$  points in the correct convolution. So, use 512 point DFTs instead.
- **Why is it called circular?** If you consider a single period, it appears that the right side of the signal is somehow connected to the left side.



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